

Homework (Carolina Reaper)

- Q1.** A musician is mixing two audio signals. The frequency of the first signal is given by $x^2 + 3x - 2$, and the frequency of the second signal is given by $2x + 1$. What is the intersection point of these two frequencies?
- (A) $x = -2$
 (B) $x = 1$
 (C) $x = -1$
 (D) $x = 2$
 (E) $x = 0$
- Q2.** An artist is creating a painting with a quadratic shape. The dimensions of the shape are given by $x^2 - 4x + 3 = 0$ and $y = 2x - 1$. What are the coordinates of the vertex of the quadratic shape?
- (A) (1, 1)
 (B) (2, 3)
 (C) (3, 5)
 (D) (4, 7)
 (E) (5, 9)
- Q3.** A guitarist is playing a rhythm pattern with a quadratic equation $x^2 + 2x - 6 = 0$. What is the x -intercept of this equation?
- (A) $x = -3$
 (B) $x = -1$
 (C) $x = 1$
 (D) $x = 2$
 (E) $x = 3$
- Q4.** A music producer is working with a quadratic function $y = x^2 - 4x + 4$. What is the axis of symmetry of this function?
- (A) $x = -2$
 (B) $x = -1$
 (C) $x = 0$
 (D) $x = 1$
- (E) $x = 2$
- Q5.** A drummer is playing a rhythm with a linear equation $y = 2x + 1$ and a quadratic equation $y = x^2 - 2x - 3$. What is the point of intersection of these two equations?
- (A) (1, 3)
 (B) (2, 5)
 (C) (3, 7)
 (D) (4, 9)
 (E) (5, 11)
- Q6.** A composer is working with a system of equations: $x^2 + 2y = 4$ and $2x - y = -3$. What is the solution to this system?
- (A) (1, 1)
 (B) (2, 2)
 (C) (3, 3)
 (D) (4, 4)
 (E) (5, 5)
- Q7.** A pianist is playing a melody with a quadratic equation $x^2 - 3x - 2 = 0$. What is the y -intercept of this equation?
- (A) $y = -2$
 (B) $y = -1$
 (C) $y = 0$
 (D) $y = 1$
 (E) $y = 2$
- Q8.** A music therapist is working with a client who is creating a song with a quadratic equation $y = x^2 + 2x - 3$. What is the vertex of this equation?
- (A) (-1, -4)
 (B) (-2, -3)
 (C) (-3, -2)
 (D) (-4, -1)
 (E) (-5, 0)

Open Ended Questions (FRQ)

- Q9.** A painter is creating a mural with a quadratic shape. The dimensions of the shape are given by $x^2 - 5x + 6 = 0$ and $y = 3x - 2$. Find the coordinates of the vertex of the quadratic shape and explain the significance of the vertex in the context of the mural.
- Q10.** A music producer is working with a system of equations: $x^2 + 4y = 8$ and $3x - 2y = 5$. Solve the system and interpret the solution in the context of music production. Be sure to explain the significance of the intersection point.

Chef's Tips

- Tip 1: Encourage students to use graphing calculators to visualize the quadratic equations.
- Tip 2: Remind students to check their work by plugging their solutions back into the original equations.
- Tip 3: Emphasize the importance of interpreting the intersection points in the context of the given scenarios.